

CHAPTER ONE

INTRODUCTION TO MARKETING INFORMATION SYSTEM

1.1 DEFINITION

1. **Information system (IS)** - an information system is an organized combination of people hardware, software, communications networks, and data resources that collects, transforms, and disseminates information in an organization to support and improve day-to-day operations of the organization as well as support the problem solving and decision making needs of management and users.

Today's end users rely on many types of information system (IS). They might include simply manual, informal communications channels or sophisticated computer based system.

1. **Data** - are raw facts or observations, typically about physical phenomena or business transactions. More specifically, data are objective measurements of the attributes or characteristics of entities (such as people, places, things and events.) Names, quantifies, and Birr amounts recorded on sales forms represent data about sales transactions.
2. **Information** - is defined as data that have been converted into a meaningful, and useful context for specific end users. Thus, data are usually subjected to a value added process. This process is known as information processing or data processing in which;
 - i. The data form is aggregated, manipulated, and organized.
 - ii. The content of the data is analyzed and evaluated; and
 - iii. The data is placed in a proper context for a human user.

For example, data contained in sales records can give meaningful sales information after it is properly organized and manipulated such as sales by product type, sales territory, or sales person. Information is organized and processed data, which is capable of communicating knowledge and meaning. Relevant information is the one that increase knowledge and reduce uncertainty.

Information to be valuable or useful and meaningful, it should fulfill some attributes or characteristics which can be expressed in terms of three dimensions of time, content, and form.

➤ *Content dimension includes*

- **Accuracy:** - Information should be free errors. In other words, the correctness of the input data and that of the processing rules should be ensured so that the resulting information is accurate. Incorrect information is worse than no information.
- **Relevance:-** For information to be relevant should be tailored to the needs of the user. For example massive volumes of irrelevant information would waste a lot of manager's time and there is a danger of his missing important relevant information.
- **Completeness** – The information should be complete. That is, it should include all data and not exclude some.

- *Conciseness* – It is essential to give brief summarized information to ensure quick action.
- *Performance*
- *Form Dimension:* This dimension of a good information include.
 - Clarity, Details, Order, Presentation and Media.
- *Time Dimension:* An information to be useful should satisfy the following characteristics: Timeliness, Currency, Frequency and Time Period

In general all information to support intelligent decision-making at all three levels in the management hierarchy must be correct; that is accurate and complete, including all relevant data; cost effective, meaning effectively obtained; yet understandable. It must be current; meaning timely; yet also time sensitive, based on historical, current, or future information needs. Thus information has the following terms and terminologies.

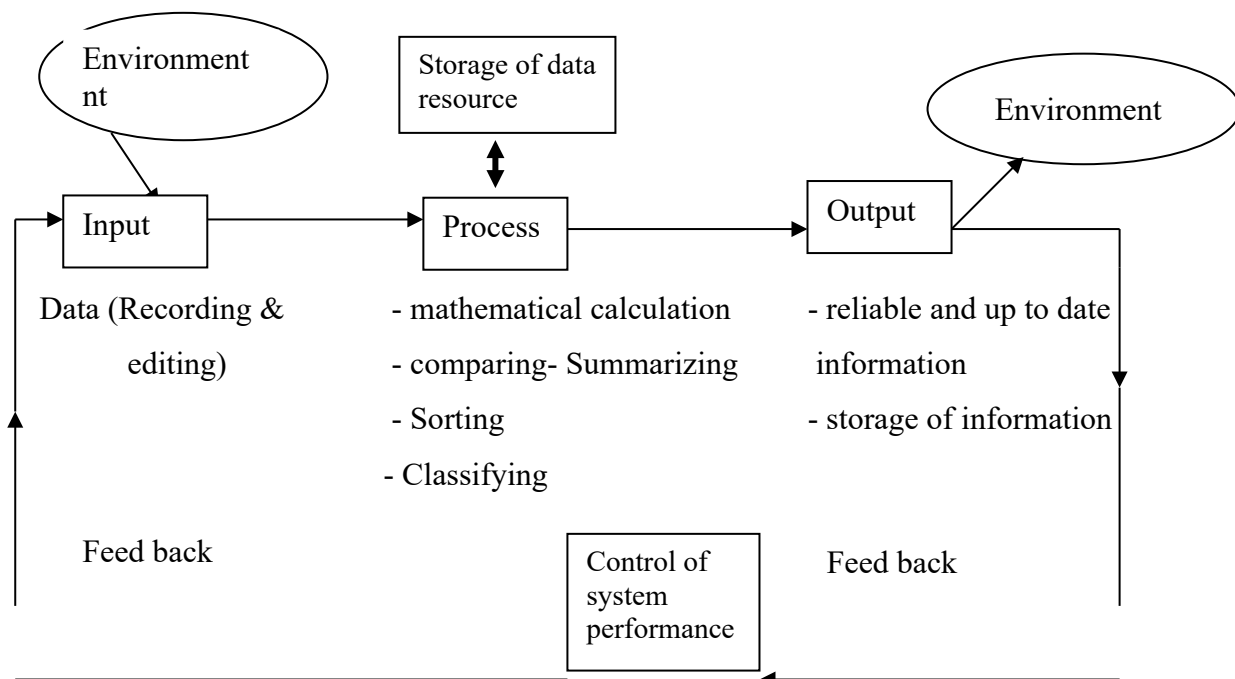
- **End Users** - are people who use an information system or the information it produces. They can be accountants, sales persons, engineers, clerks, customers, or managers. It includes anyone who uses an information system or the information except few numbers of people who are known as information system specialists (system analysts, programmers, and computer engineers etc).
- **Information Technology (IT)** - is a contemporary term that describes the combination of computer technology (hardware and software) with telecommunications technology (data, image, and voice networks).
- **Organization**- is a collection of people working together to achieve a common purpose, with that purpose being to produce goods and/or services that satisfy the needs of customers. An organization is a system that interacts with an environment. It takes resources (inputs) from the environment and processes them to produce out puts and supply it to the environment using its basic operational functions like purchasing and supplies management, marketing, production and finance. This cycle is given by the diagram below.
- **Marketing.** The American Marketing Association (AMA) defined marketing as “the process of planning and executing the conception, pricing, promotion, and distribution of ideas, goods, and services to create exchanges that satisfy individual and organizational objectives. It is the performance of business activities that direct the flow of goods and services from producers to consumers or users. According to Clerk, “marketing consists of those activities which effect transfers in ownership of goods and care for their physical distribution.”

What is System?

A **system** can be defined as a group of interrelated or interacting elements forming a unified whole. Examples of system include the physical system of the sun and its planets, the biological system of the

human body, the technical system of a textile factory, and the socioeconomic system of business organization. A system is a group of interrelated components working together toward a common goal by accepting inputs and producing outputs in an organized transformation process. There are large numbers of systems are around as such as the biological system, the physical system, the socio-economical system etc.

Information system is one of the several systems and has the following basic interacting components or functions as given in the model below.



Following is a brief description of these components.

1. **Input** - involves capturing and assembling data elements that enter the system to be processed by data entry activities such as recording and editing.
2. **Processing** - involves *transformation processes* that convert input data into output (Information). It involves manipulating data by activities of calculating, comparing, sorting, classifying and summarizing.
3. **Output** - involves *transforming data* that have been produced by a transformation process to their ultimate destination, that is, end users. This can be shown diagrammatically as follows
4. **Storage of data resources**: this is the activity in which data and information are retained in an organized manner for later use. Normally stored data is organized into fields, records, files and data bases.
5. **Control and feedback of system**: An information system is expected to produce feedback about its input, processing output and storage activities. Feedback is a data about the performance of a

system. The feedback must be monitored and evaluated to determine if the system is meeting established performance standards.

2. **Information system Resources:** In order to perform the different activities described above, MIS requires certain resources. The basic ones are:

- i. **People resource.** This includes both end users and IS specialists. End users are people who use information system after they have been trained, sensitized and educated to use the information produced from the system. End users can be accountants, sales persons, engineers, customers, or managers. IS specialists are people who develop and operate the ISs. They include system managers, computer operators, system analysts, and programmers.
- ii. **Hardware resource.** The computer systems, peripherals and telecommunication equipment which are needed in order to perform the processing, storing and communication of the information.
- iii. **Software resource.** Different instructions that interact with the hardware to function and perform its tasks. The software resources could be system software resources and application software resources.
- iv. **Data resources.** Depending on the complexity of the system the data resources of the information system are typically organized into database, model base, and knowledge base. Database hold processed and organized data and knowledge base holds knowledge in a variety of forms such as facts, rules, and case examples about successful business practice. For example, data about sales transaction may be accumulated and stored in a sales database frequent processing that yields daily, weekly, monthly and yearly sales analysis report management.

The systems concept also considers the organization as a system, which interacts with its environment (open system) by taking inputs and giving its outputs to the environment.

3. The Role of Information System

An information system plays the following roles in business operation.

1. The functional support role

The business processes and operations support function is the most basic. It involves collecting, recording, storing, and basic processing of data. Information systems support business processes and operations by:

- recording and storing sales data, purchase data, investment data, payroll data and other accounting records
- processing these accounting records into income statements, balance sheets, ledgers, management reports, and other forms of financial information

- recording and storing inventory data, work in process data, equipment repair and maintenance data, supply chain data, and other production/operations records
- processing these operations records into production schedules, production controllers, inventory systems, and production monitoring systems
- recording and storing personnel data, salary data, employment histories, and other human resources records
- processing these human resources records into employee expense reports, and performance based reports
- recording and storing market data, customer profiles, customer purchase histories, marketing research data, advertising data, and other marketing records
- processing these marketing records into advertising elasticity reports, marketing plans, and sales activity reports
- recording and storing business intelligence data, competitor analysis data, industry data, corporate objectives, and other strategic management records
- processing these strategic management records into industry trends reports, market share reports, mission statements, and portfolio models
- use of all the above to implement, control, and monitor plans, strategies, tactics, new products, new business models or new business ventures

3.The decision support role

The business decision making support function goes one step further. It is an integral part of making decisions. It allows users to ask “What if . . . ?” questions: What if we increase the price by 5%? ; What if we increase price by 10%? ; What if we decrease price by 5%? ; What if we increase price by 10% now, then decrease it by 5% in three months? It also allows users to deal with contingencies: If Inflation increases by 5% (instead of 2% as we are assuming), then what do we do? What do we do if we are faced with a strike or a new competitive threat?

The most basic and most versatile business decision making tool is the spreadsheet, but spreadsheets are not user friendly. More sophisticated programs often seamlessly incorporate statistical decision making tools like sensitivity analysis, Monte Carlo analysis, risk analysis, break even analysis and Bayesian analysis.

3.The strategic support role

Information systems can support a company's competitive positioning. Here are three levels of analysis:

- i. ***The supports for help in piloting the chain of internal value.*** They are the most recent and the most pragmatic systems within the reach of the manager. They are the solutions to reductions of costs and management of performance. They are typically named "Business Workflow Analysis" (BWA) or of "Business Management Systems".
- ii. ***All successful companies have one (or two) business functions that they do better than the competition.*** These are called core competencies. If a company's core competency gives it a long term advantage in the marketplace, it is referred to as a sustainable competitive advantage. For a core competency to become a sustainable competitive advantage it must be difficult to mimic, unique, sustainable, superior to the competition, and applicable to multiple situations. Examples of company characteristics that could constitute a sustainable competitive advantage include: superior product quality, extensive distribution contracts, accumulated brand equity and positive company reputation, low cost production techniques, patents and copyrights, government protected monopoly, and superior employees and management team. The list of potential sustainable competitive advantage characteristics is very long. However, there are some commentators' claims that in a fast changing and competitive world, none of these advantages can be sustained in the long run. They argue that the only truly sustainable competitive advantage is to build an organization that is so alert and so agile that it will always be able to find an advantage, no matter what changes occur.
- iii. ***Information systems often support and occasionally constitute these competitive advantages.*** The rapid change has made access to timely and current information critical in a competitive environment. Information systems, like business environmental scanning systems, support almost all sustainable competitive advantages. Occasionally, the information system itself is the competitive advantage.

4. The performance monitoring role

MIS are not just statistics and data analysis. They have to be used as a MBO / Management by Objectives/ tool. They help:

- to establish relevant and measurable objectives
- to monitor results and performances (reach ratios)
- to send alerts, in some cases daily, to managers at each level of the organization, on all **deviations** between results and pre-established objectives and budgets.